Step 1: In the Seg_dwi package, a DWI example and the segmentation code are provided. Simply place the appropriate data or modify the file path accordingly, and the segmentation mask will be generated in the /output folder.

The segmentation code used in this study is proprietary and not open-source, as it involves internal code from United Imaging. The code uploaded here is the segmentation pipeline from our previous work.

Step 2: Using the refine_aspect_template_mca333_extend.nii atlas, run the cal_lesion_load_thresholds script to obtain region-specific thresholds for the 20 ASPECTS regions.

Ten examples are provided here to facilitate the calculation of lesion load thresholds.

Run the script using the following command:

Bash

```
python "cal_lesion_load_thresholds.py" --input ratio.xlsx
```

Step 3: Input the calculated thresholds into the cal_Ref_AI_ASPECTS code to compute the Ref-AI-ASPECTS score for each patient based on the predefined thresholds.

For demonstration, a sample file (STK_CZ1_DWI_00003.nii) is provided. To compute the score, run the following command in the command line:

Bash

```
python cal_Ref_AI_ASPECTS.py --mask STK_CZ1_DWI_00003.nii --atlas refine
_aspect_template_mca333_extend.nii
```

where --mask specifies the path to the lesion mask to be evaluated, and --atlas specifies the path to the atlas template.